Scientific Communication

Writing, slide design and presentation skills

A three-hour practical workshop

Neudata

Insight. Impact. Innovation.

You will leave with four things you made yourself



Green: modules with practice. Grey: welcome. Pale: 12-minute break.

Revised paragraph

Claim, evidence and caution in the right order

Storyboard

A short talk planned as an argument

Redesigned slide

Headline plus evidence

Rehearsed explanation

Two timed minutes, with feedback

You also take away a workbook, worksheets, a rehearsal checklist and a feedback rubric.

We give feedback on clarity and understanding, never on accent or personality

Feedback on the work

Clarity, evidence and understanding. Never accent, looks or personality.

No confidential data

Fictional cases, or your own work with identifiers removed.

What is said here stays here

Share ideas freely, including unfinished ones.

Practice, not performance

Expect rough drafts. Nothing today is graded.

One fictional study runs through the session, told to three different rooms

850
patients analysed, two hospitals

14.6%
readmitted with a call (60 of 412)

21.5%
readmitted with no call (94 of 438)

Calls were **not randomised**. Nurses called when free, and only patients who could be reached.

The three rooms

Conference
Researchers presenting to peers

Thesis defence
Students facing examiners

Hospital or organisation
Professionals briefing decision-makers

A second fictional case, CoolRoom, covers classroom temperature for non-clinical colleagues.

Fictional teaching data, not a real study.

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Audience, purpose and message

Module 1: 20 minutes

Every talk starts with people who must do something afterwards



Work left to right. If you cannot fill a box, you are not ready to make slides.

Teaching recommendation, drawing on Bourne 2007 and the CDC Clear Communication Index (professional guidance).

A topic names an area; a message makes a claim someone could challenge

TOPIC

Telephone follow-up after hospital discharge

MESSAGE

A nurse call is linked to fewer readmissions, but the evidence does not yet justify full roll-out

Three checks for your sentence

Is it a full sentence?

It has a verb and says something happened.

Could someone disagree?

If nobody could, it is still a topic.

Could you say it in a corridor?

It should work without slides.

Fictional teaching data, not a real study. neu-data teaching recommendation.

One finding needs a different message in each room

Room	Who they are	What they should do	Your message
Conference audience	Peers in the field, mixed methods knowledge	Consider a new explanation and ask how it applies to their setting	In two hospitals, a nurse call was linked to 6.9 fewer readmissions per 100, but much of that gap reflected who could be reached.
Thesis committee	Examiners who know the field and will probe your choices	Judge that the work is yours, sound and a contribution	I show that reachability confounds observational evaluations of follow-up calls, and I defend the stratified analysis that reveals it.
Hospital management	Decision-makers with budgets, little time, limited methods detail	Decide what to do next	We recommend a six-month staggered trial on two wards before funding full roll-out, because the benefit could be 9.4 fewer to 0.8 more readmissions per 100.

Fictional teaching data, not a real study. neu-data teaching examples.

Fill in Worksheet 1 for your own talk

Activity, 13 minutes. Worksheet 1 alone (10), then pair check (3). Workbook page: Worksheet 1.

Alone, 10 minutes

Name your audience and what they know

State what should change afterwards

Write your message in one sentence

In pairs, 3 minutes

Read your sentence aloud

Partner says back what they think you want them to do

Revise if it does not match

No current talk? Use CareCall or CoolRoom and pick one room.

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Scientific writing

Module 2: 30 minutes

A clear paragraph makes one point, then supports it

Point	Patients who got a nurse call were readmitted less often.
Evidence	Readmission within 30 days was 14.6% with a call and 21.5% without (60 of 412 and 94 of 438).
Interpretation	The gap may partly reflect who was reached, as high-risk patients received fewer calls.
Implication	A randomised evaluation is needed before roll-out.

Within a sentence: start with what the reader knows, and end with what is new.

Fictional teaching data, not a real study. Teaching resource: Gopen & Swan 1990 (argued essay, not rules). Example: neu-data.

Most unclear sentences are vague, wordy or full of needless jargon

Problem	Before	After
Vague	Readmissions were considerably lower in the call group.	Readmissions were 6.9 percentage points lower in the call group (14.6% v 21.5%).
Wordy	Due to the fact that a large number of patients could not be contacted, it was not possible to make calls to all of them.	Nurses could not call every patient, so 438 received no call.
Jargon	Post-discharge telephonic engagement modalities were operationalised by nursing staff.	Nurses telephoned patients after discharge.

Fictional teaching data, not a real study. Professional guidance: Federal Plain Language Guidelines; digital.gov plain language guide.

Keep what you found separate from what you think it means

RESULT: what the data show

60 of 412 patients with a call (14.6%) were readmitted, and 94 of 438 without (21.5%).

29% of those called were at high risk, against 45% of those not called.

INTERPRETATION: what you think it means

The call **may** have helped some patients.

The gap **may also reflect** who could be reached, because healthier patients were easier to contact.

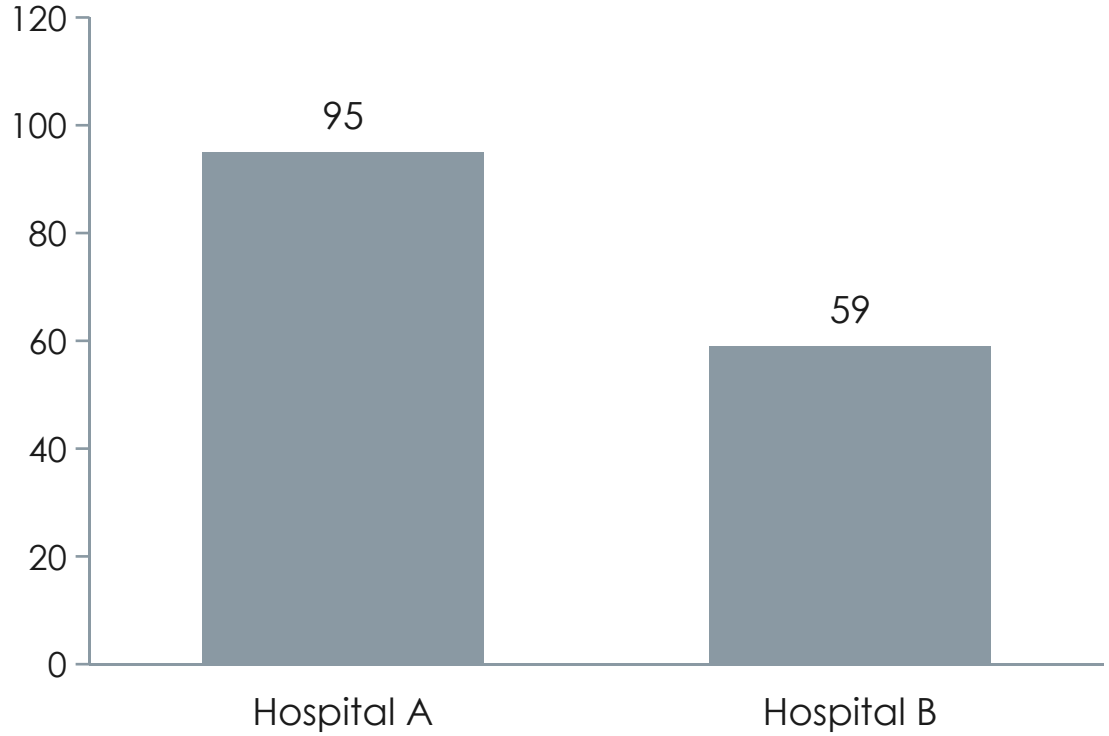
Avoid mixing them

"The calls worked, with only 14.6% readmitted."

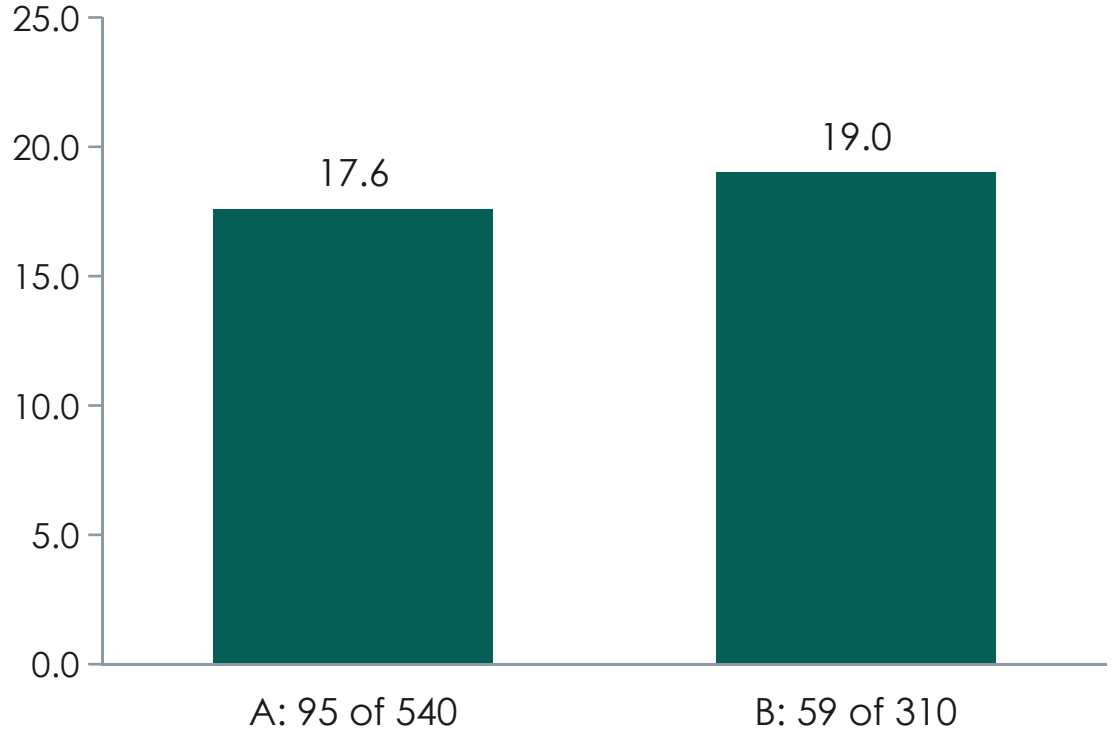
Fictional teaching data, not a real study. Teaching resource: Manchester Academic Phrasebank. Guidance: STROBE.

A count without a denominator can reverse the picture

Counted: Hospital A looks worse



As a rate: Hospital B is higher



Give numerator and denominator every time: 95 of 540, not just 95 and not just 17.6%.

Fictional teaching data, not a real study. Reporting guidance: SAMPL (Lang & Altman 2015).

Report the size of the difference and how uncertain it is

32%

relative reduction alone: sounds large, hides the base

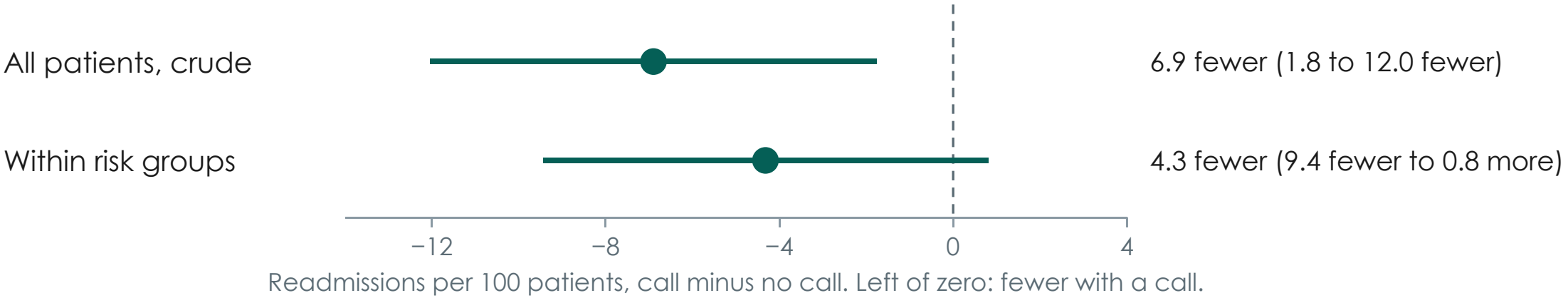
6.9 per 100

absolute difference: 21.5% v 14.6%

1.8 to 12.0

95% confidence interval: the range the data fit

Difference in readmissions



Fictional teaching data, not a real study. Professional guidance: SAMPL, STROBE. Research finding: Akl et al. 2011 (Cochrane), van der Bles et al. 2020.

Choose the verb your design can support

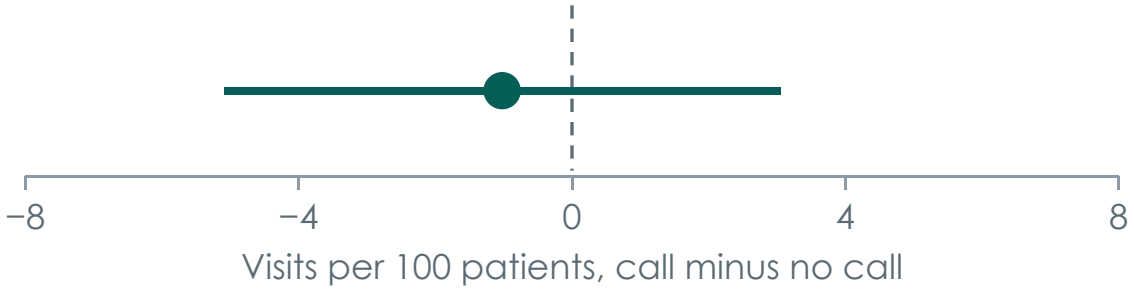
What you have	Verbs that fit	Too strong for it
Observational comparison, as in CareCall	was associated with, was higher among	led to, reduced, prevented
Comparison with adjustment	remained lower after stratifying by risk	the effect of, independently caused
Well-conducted randomised comparison	reduced, in this population	proves, works everywhere
Any single study, in the conclusion	may, suggests, is consistent with	shows conclusively, will improve

The word "association" is not a shield. Many readers still hear a causal claim, so state the design and the main alternative explanation as well.

Research findings: Haber 2018, 2022. Guidance: STROBE, GRADE 26 (for reviews). Wording ladder: neu-data teaching recommendation.

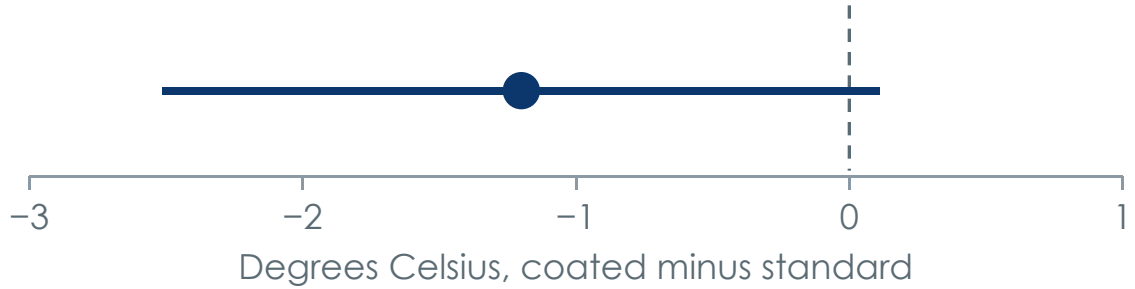
A non-significant result is rarely proof of no effect

CareCall: emergency visits



1.0 fewer per 100, but anything from 5.1 fewer to 3.0 more fits the data.

CoolRoom: classroom temperature



1.2 °C cooler, with 2.5 °C cooler to 0.1 °C warmer fitting the data.

Do not write: "The calls had no effect."

Write: "The study cannot tell us whether calls change emergency visits. The interval includes a useful benefit and a small harm."

Fictional teaching data, not a real study. Review: Altman & Bland 1995, Greenland et al. 2016, ASA statement 2016.

Rewriting made this result longer, and much more honest

BEFORE

It is a well-known fact that post-discharge follow-up is of great importance. The implementation of nurse telephone calls in our hospitals led to a highly significant reduction in readmissions of 32% ($p<0.05$), which proves that the intervention works and should be rolled out everywhere. There was no effect on emergency visits.

WHAT WE CHANGED

Removed the opening claim

Replaced the verb

Added counts and denominators

Reported the absolute difference and its interval

Exposed the confounder

Moved the recommendation out

Fictional teaching data, not a real study. Worked example: neu-data. Full rewrite in the workbook.

One finding takes four forms, and the strength of the claim stays the same

Paragraph	Abstract passage	Slide headline	Spoken explanation
139 words	73 words	16 words	57 words
Full reasoning for a reader who can reread.	Result, adjusted result and a cautious conclusion.	Patients who got a call had fewer readmissions, but the gap narrowed after accounting for risk	"For every hundred patients, those who got a call had about seven fewer readmissions..."

Keep in every version: the size of the difference, who was studied, that calls were not randomised, and that the result is uncertain. Words can go. These cannot.

Fictional teaching data, not a real study. Worked examples: neu-data. Abstract structure: MIT Communication Lab; STROBE.

Edit the CoolRoom paragraph, or a paragraph of your own

Activity, 12 minutes. Worksheet 2. Alone (8), then swap and comment (4).

In light of the fact that excessive heat in classrooms constitutes a significant problem for learning, the implementation of a reflective roof coating was undertaken in a number of classrooms. It was found that temperatures were much lower in the coated rooms, showing that the coating cools classrooms and will improve exam results. However, the difference was not statistically significant ($p>0.05$), so it can be concluded that the coating does not work.

Facts you may use

12 coated and 12 standard rooms

29.4 °C and 30.6 °C on average

95% CI of the difference: 2.5 °C cooler to 0.1 °C warmer

Head teachers chose the rooms

Exam results not measured

Fictional teaching data, not a real study.

A strong edit reports the data, names the limits and drops the unsupported claims

We asked whether a reflective roof coating lowers classroom temperature. Mean afternoon temperature was 29.4 °C in 12 coated classrooms and 30.6 °C in 12 standard classrooms, a difference of 1.2 °C (95% CI 2.5 °C cooler to 0.1 °C warmer). The interval includes no difference and a useful cooling, so this small pilot cannot settle the question. Head teachers chose the coated rooms, so other differences between rooms could explain some of the gap. We did not measure exam results.

Removed: wordy opening, "significant" for importance, "will improve exam results", "does not work".

Added: sizes of groups, means, difference, interval, how rooms were chosen.

Fictional teaching data, not a real study. Model answer: neu-data. Other good versions exist.

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Presentation structure

Module 3: 20 minutes

A talk is one argument: question, evidence, interpretation, implication

Question	Evidence	Interpretation	Implication
Are nurse calls linked to fewer readmissions?	14.6% v 21.5%. Within risk groups, 4.3 fewer per 100.	A signal, partly explained by who was reached.	Test on two wards before full roll-out.

If a slide serves none of the four boxes, cut it.

Fictional teaching data, not a real study. neu-data teaching recommendation.

Open with the problem and the question, not with an outline

A WEAK OPENING

"Good morning. Thank you for having me. Today I will first give some background, then the methods, then results, discussion and conclusion..."

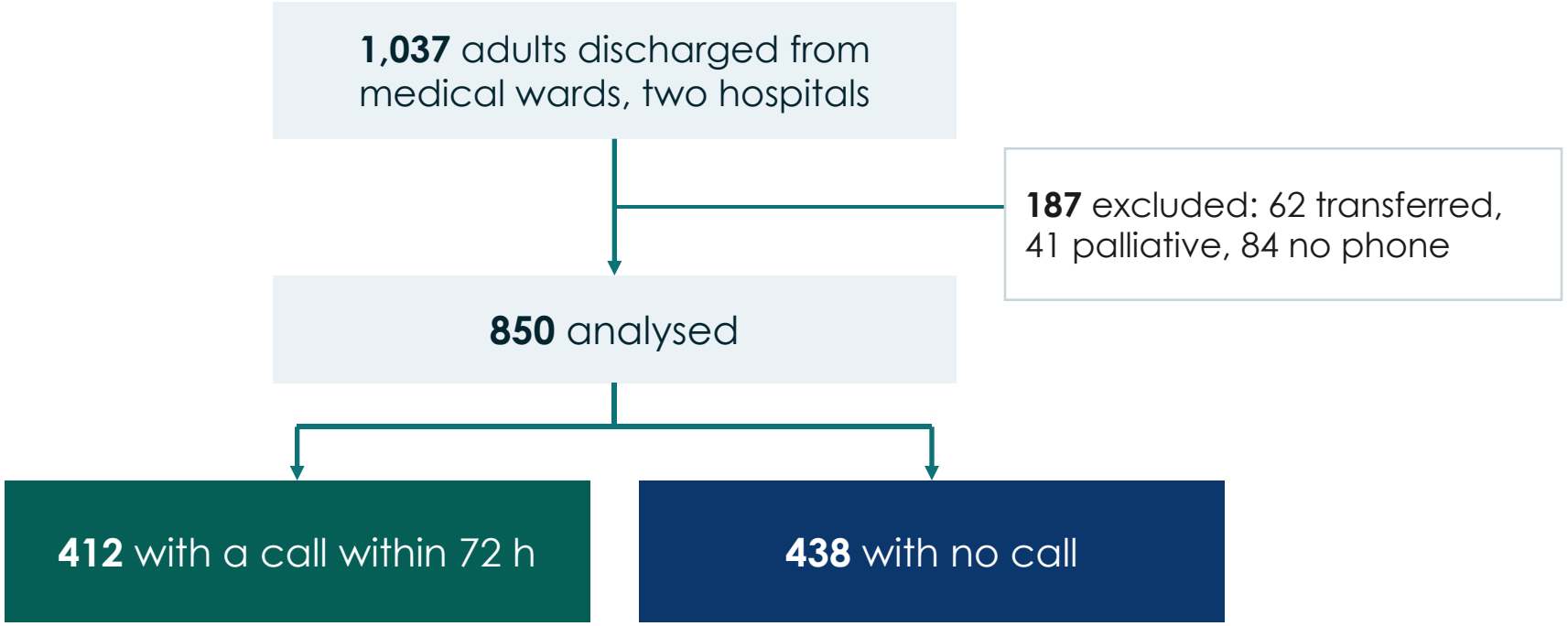
A STRONGER OPENING

"Last year, about 18 in every 100 patients on our medical wards came back within a month. Nurses started phoning some of them after discharge. We asked: did those calls make a difference?"

Problem, then question. Then a one-line roadmap if the talk is long.

Fictional teaching data, not a real study. Teaching recommendation, consistent with Bourne 2007 and Naegle 2021 (professional advice).

Show only the methods your audience needs to judge the evidence



Four things to show

Who

What was compared

What was measured

Main threat to validity

Everything else goes to backup slides, ready for questions.

Fictional teaching data, not a real study. neu-data teaching recommendation. Backup slides: Naegle 2021.

Lead with your main finding, then give the limits that change what it means

Priority	CareCall	Time
Main finding	6.9 fewer readmissions per 100 with a call	Most
Limit that changes the reading	Within risk groups, 4.3 fewer, and the data fit 9.4 fewer to 0.8 more	Most
Secondary finding	Emergency visits: too uncertain to say	Brief
Detail	Hospital breakdown, sensitivity checks	Backup

Fictional teaching data, not a real study. neu-data teaching recommendation.

The room changes where you start and where you spend your time



Illustrative proportions: neu-data teaching suggestion, not a finding. Adjust to your setting.

A defence examines your contribution and your choices, so check your own rules first

Format example	Public?	Talk?
UK viva (Imperial)	No	Not required by the procedure
US defence (UCLA Neuroscience)	Yes, then closed session	45 to 60 minutes
Sweden (Halmstad)	Yes, with opponent	About 20 minutes, in most fields

Prepare for any format

Contribution: what is new, and what is yours

Choices: why this method and not the alternatives

Backup slides: expected challenges, tables, sensitivity results

CareCall example: "I chose stratification over a simple comparison because reachable patients differ. A randomised design was not possible for this retrospective data."

Institutional examples: Imperial College, University of Minnesota ECE, UCLA Neuroscience, Halmstad University. Each covers one institution.

End with what the evidence supports, not with "thank you"

Conference	Thesis defence	Hospital briefing
Calls were linked to fewer readmissions, but much of the gap may reflect who was reached. **Test it in a randomised design.**	This thesis shows why reachability must be handled in evaluations of follow-up calls. **Next: a design that removes it.**	We recommend a six-month staggered trial on two wards. **Decision needed today.**

Claim, caution, next step. Then stop.

Fictional teaching data, not a real study. Teaching recommendation, consistent with Naegle 2021 (professional advice).

Draft your storyboard on six cards

Activity, 12 minutes. Worksheet 3. Storyboard (10), then read your headlines to a partner (2).

1 Problem and question	2 What we did	3 Main finding
4 Limit or second finding	5 What it means	6 Conclusion or recommendation

Write each headline as a full sentence. Sketch the evidence below it.

Title test: read the headlines in order. Do they tell the story alone?

Back in 12 minutes

Next: making slides that show your evidence.

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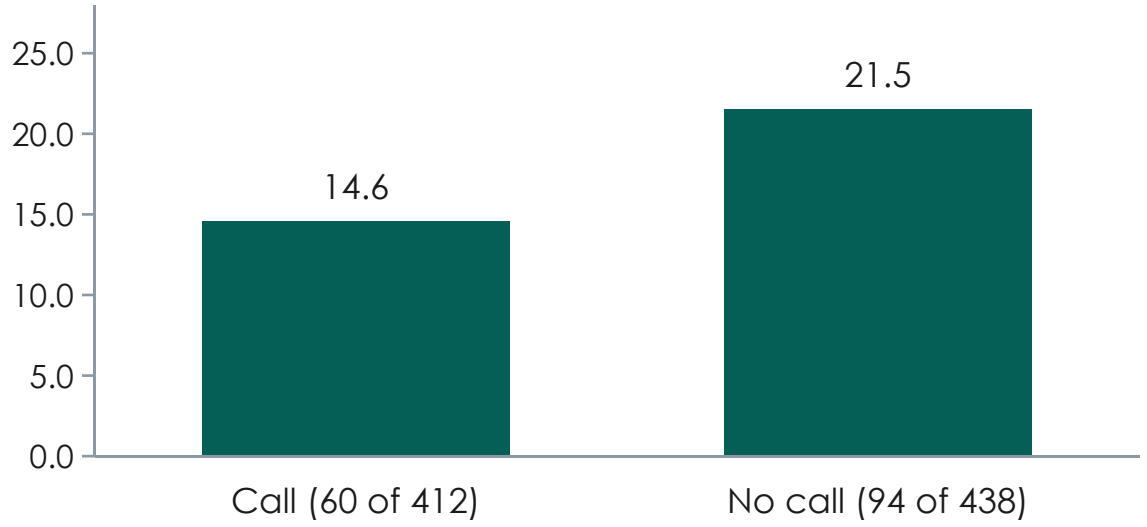
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Slide design and data communication

Module 4: 34 minutes

A slide headline states the claim; the body shows the evidence

Readmissions were lower among patients who received a call



Not randomised. Fictional data.

Headline

A full sentence that makes the claim.

Evidence

A chart, diagram, photo or table that supports it.

Spoken detail

You explain. Secondary detail goes in your notes.

What the evidence shows

In one experiment this bundle beat topic headlines with bullets. It has critics, and no study tests each part alone. Treat it as a strong default.

Research finding: Garner & Alley 2013 (a bundle of changes, recorded talk, students). Critique: Farkas 2010. Method: Alley, Penn State.

Slide-count, word-count and font-size rules are rules of thumb, not findings

Rule you may hear	What sources actually say	Use it like this
One slide per minute	Bourne: more than one visual a minute is too many, as his rule of thumb. Naegle: a minute or less. Other tested talks ran faster.	A planning check. A dense slide needs longer; a build needs less.
Six words, six lines	No primary source found. Advice varies. The direction (less text while you speak) has reasoned support.	Ask what must be read, and cut the rest to your notes.
Minimum 24 or 30 point	Published advice ranges from 18 to 32 pt. Legibility depends on screen size and distance.	Stand at the back of the actual room, or test at reduced size.

Our search found no experiment setting these numbers. Sources in the notes. Guidance labelled as such.

Readable means tested: type, spacing, contrast and layout

Contrast is measurable

Heading teal on white	**5.6:1** passes
Dark grey on white	**5.5:1** passes
Brand mid-grey on white	**2.9:1** fails
White on navy	**16.2:1** passes
Navy on mist	**14.3:1** passes

Level AA asks for 4.5:1, or 3:1 for large text. Check in the actual room too.

Four habits that cost nothing

One typeface family.
Two sizes for text, one for headlines.

Same place, every slide.
Headline, evidence, source.

Space separates ideas.
Use it before you use boxes.

Back-of-room test.
Can a colleague read it from the last row?

Standard: WCAG 2.2 SC 1.4.3 (web; W3C notes apply it to documents). Guidance: W3C Making Events Accessible.

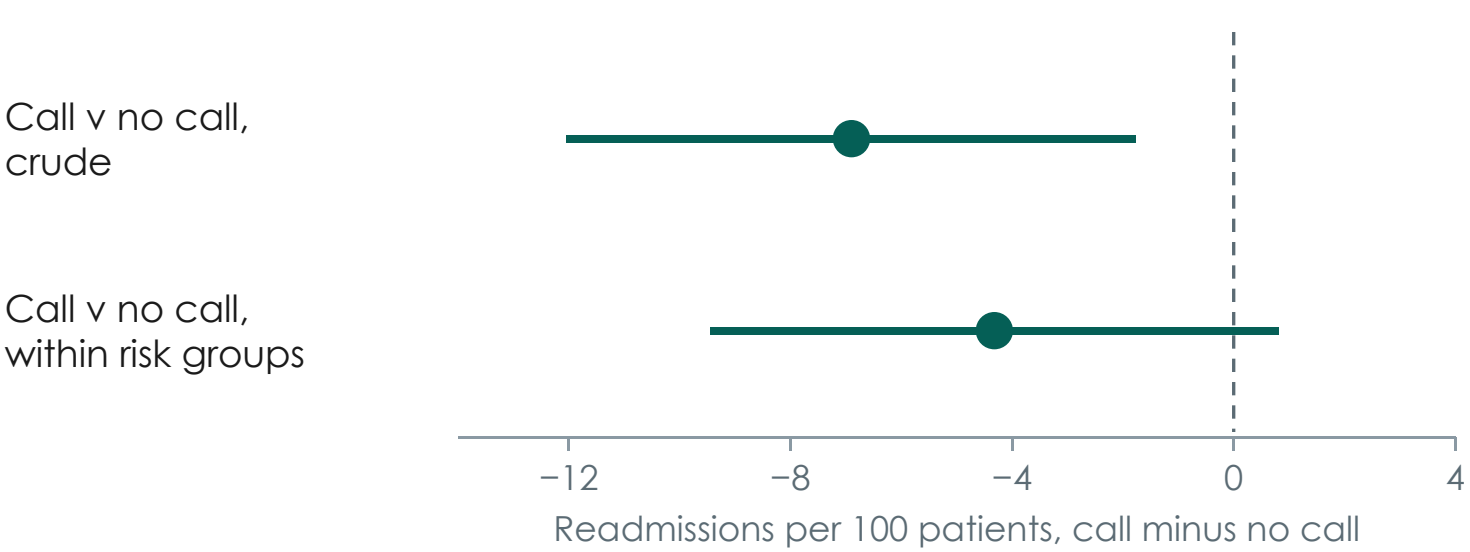
Choose the chart from the question you are asking

Your question	Good default	Be careful
How do groups compare?	Bars from zero, or dots with intervals	Bars hide spread in small continuous data
How does it change over time?	Line, with real time spacing	Short windows can flatter a trend
How spread out are values?	Individual points, box plot	A mean alone hides the shape
How do two measures relate?	Scatter plot	A pattern is not a cause
What share of the whole?	Ordered bars; stacked bar for few parts	Readers judge angle and area poorly

In perception experiments, people judged position better than length, and length better than angle or area.

Research finding: Heer & Bostock 2010, Weissgerber 2015. Guidance: Wilke 2019. Table: neu-data teaching recommendation.

Every data figure must answer five questions for the audience



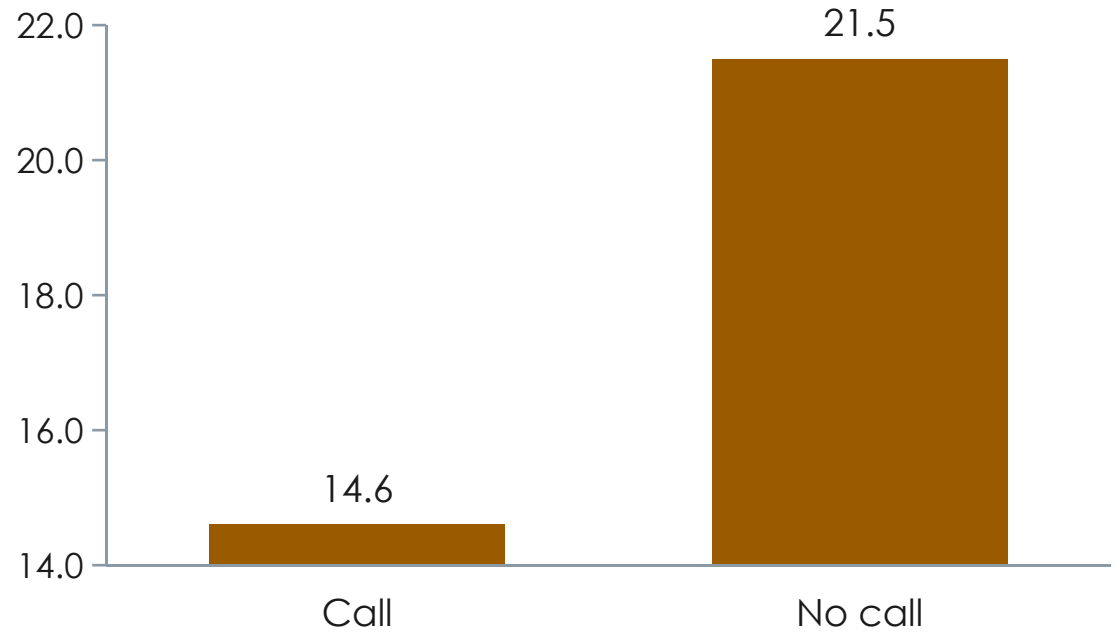
Adults discharged from medical wards at two hospitals, n = 850. Bars show 95% CI.
Source: fictional CareCall data.

1. **Axes**
   What is on each axis?
2. **Units**
   Per 100? Degrees? Days?
3. **Denominators**
   Out of how many?
4. **Uncertainty**
   What do the bars mean?
5. **Source**
   Where did the data come from?

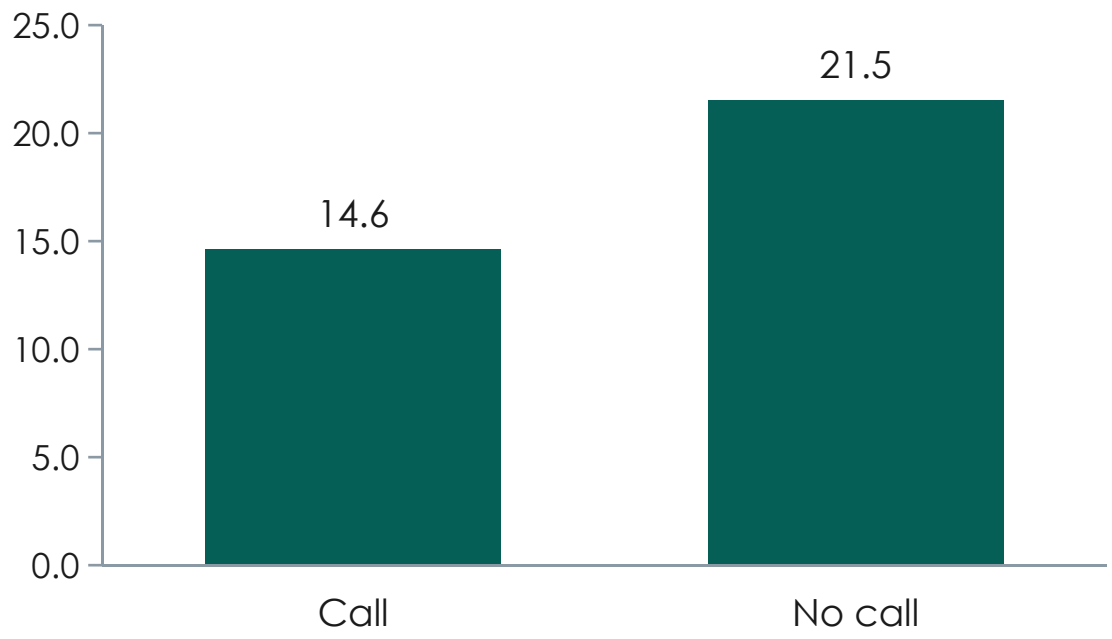
Fictional teaching data, not a real study. Teaching checklist: neu-data, drawing on SAMPL and Rougier et al. 2014.

The same numbers can look dramatic or modest, so choose honestly

Axis starts at 14: the gap looks huge



Axis starts at 0: the gap is in proportion



Bars encode length, so bars start at zero. Dots and lines encode position and may use a narrower range, if the range is labelled and the change is meaningful.

Fictional teaching data, not a real study. Research findings: Pandey et al. 2015, Correll et al. 2020 (online samples). Guidance: Wilke 2019.

BEFORE

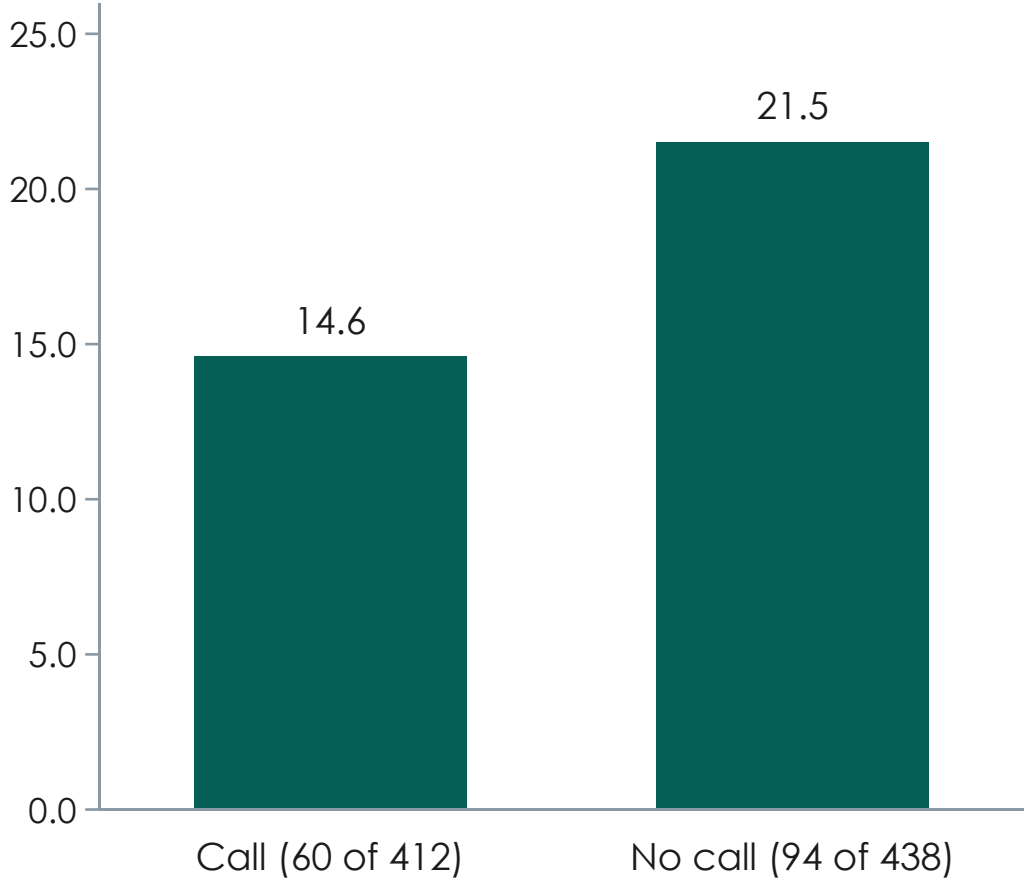
Results

- Total sample size was 850 patients discharged from medical wards at Hospitals A and B during the study period
- 412 patients received a nurse telephone call and 438 patients did not receive a nurse telephone call
- In the call group 60 patients (14.6%) had an unplanned readmission within 30 days
- In the no-call group 94 patients (21.5%) had an unplanned readmission within 30 days
- The risk difference was 6.9 percentage points (95% CI 1.8 to 12.0)
- Patients at high discharge risk made up 29% of the call group and 45% of the no-call group
- After stratification by discharge risk the difference was 4.3 points (9.4 lower to 0.8 higher)
- This suggests that the intervention may have had a beneficial impact on readmission outcomes

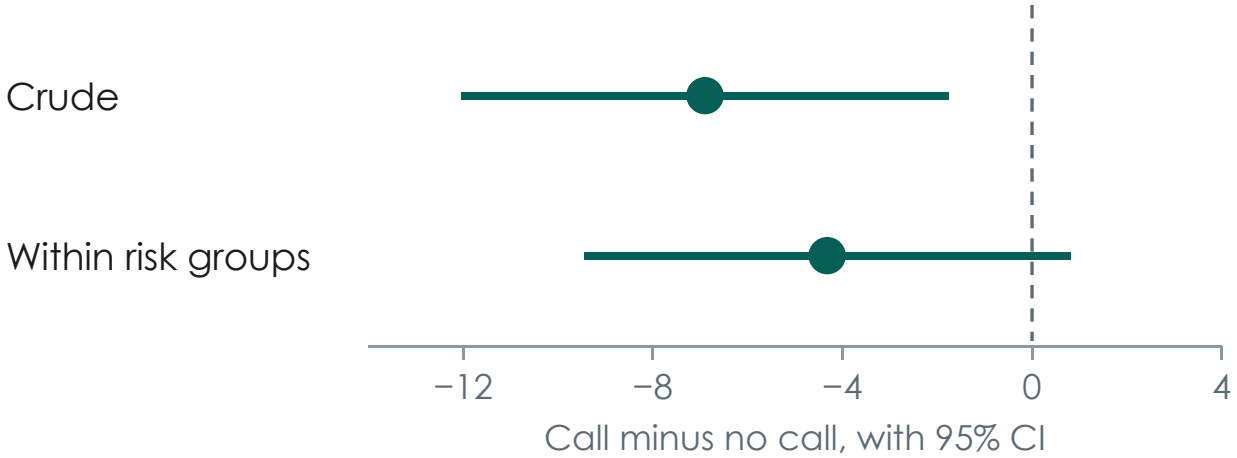
Fictional teaching data, not a real study.

AFTER

Patients who got a call had fewer readmissions, but the gap narrowed after accounting for risk



Difference in readmissions per 100 patients



High-risk patients were harder to reach: 29% of those called, 45% of those not called.

Calls were not randomised.

Fictional teaching data, not a real study.

BEFORE

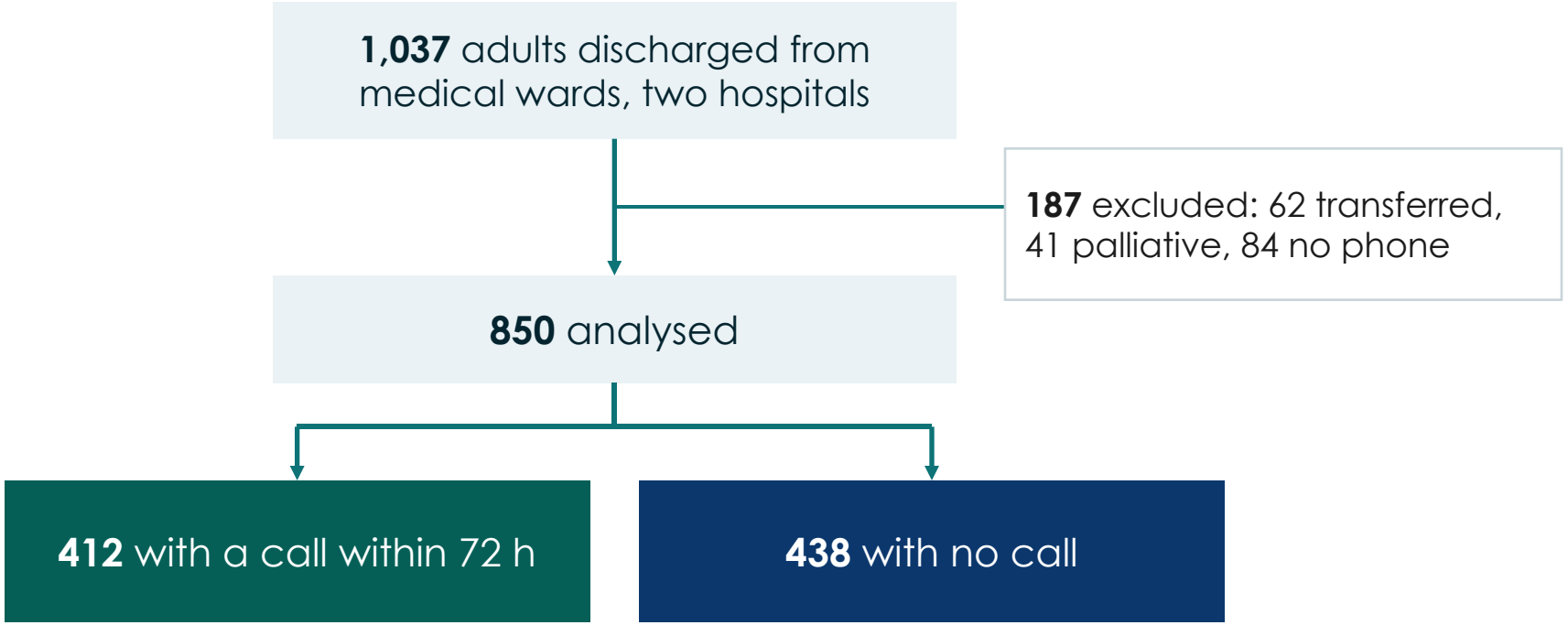
Methods

- Retrospective observational cohort study
- Two hospitals, medical wards, one calendar year
- Inclusion criteria: adults discharged home
- Exclusion criteria: transfer (n=62), palliative pathway (n=41), no working phone number (n=84)
- Exposure: nurse telephone call within 72 hours of discharge
- Outcome: unplanned readmission within 30 days
- Statistical analysis: risk differences, 95% Wald confidence intervals, Mantel-Haenszel stratification by discharge risk (high or lower)

Fictional teaching data, not a real study.

AFTER

We compared 850 patients who did or did not get a call within 72 hours



Why it matters

Nurses called when free and when a patient could be reached, so groups may differ.

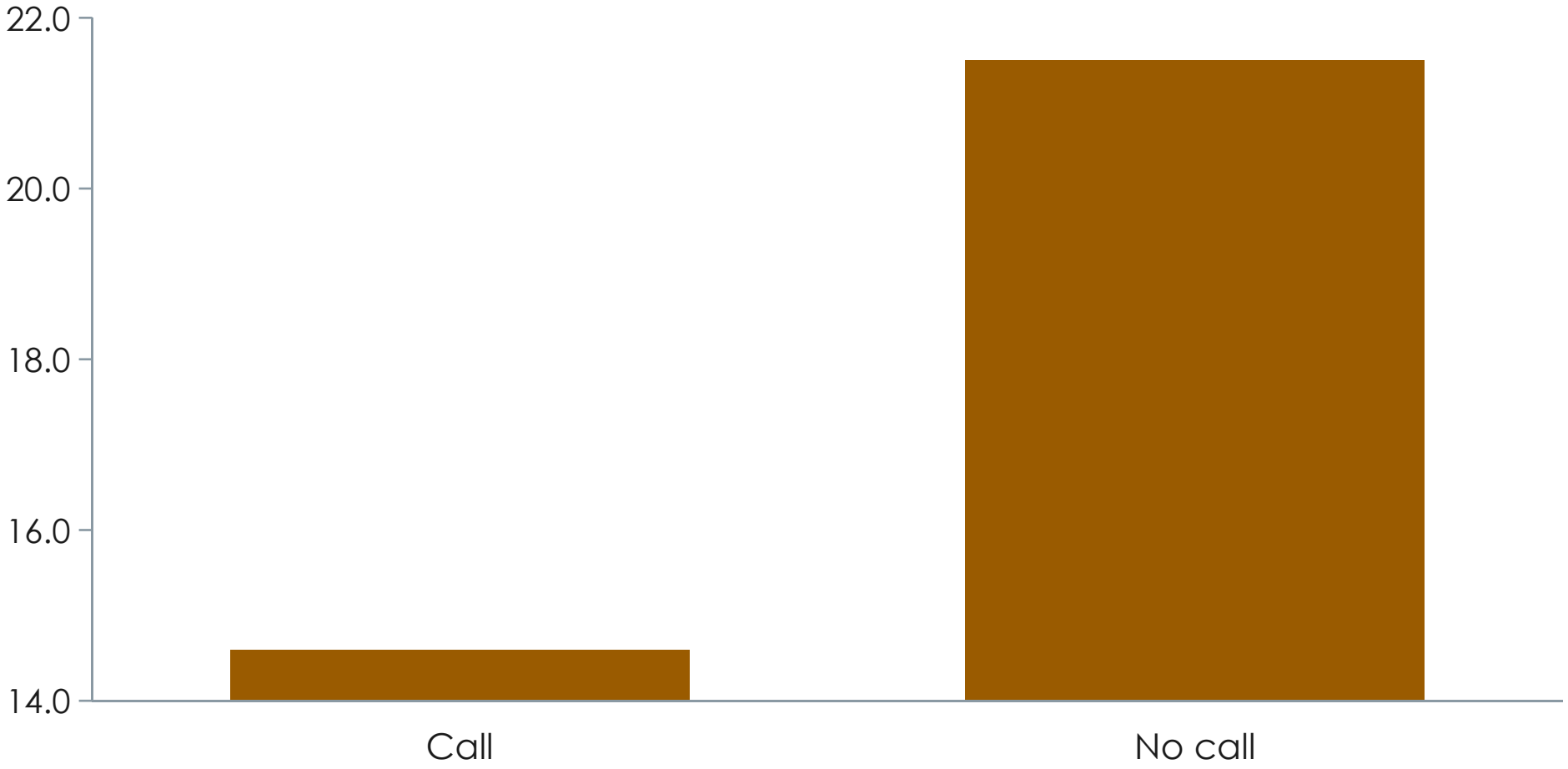
Outcome: unplanned readmission within 30 days.

Analysis: risk difference, overall and within discharge-risk groups.

Fictional teaching data, not a real study.

BEFORE

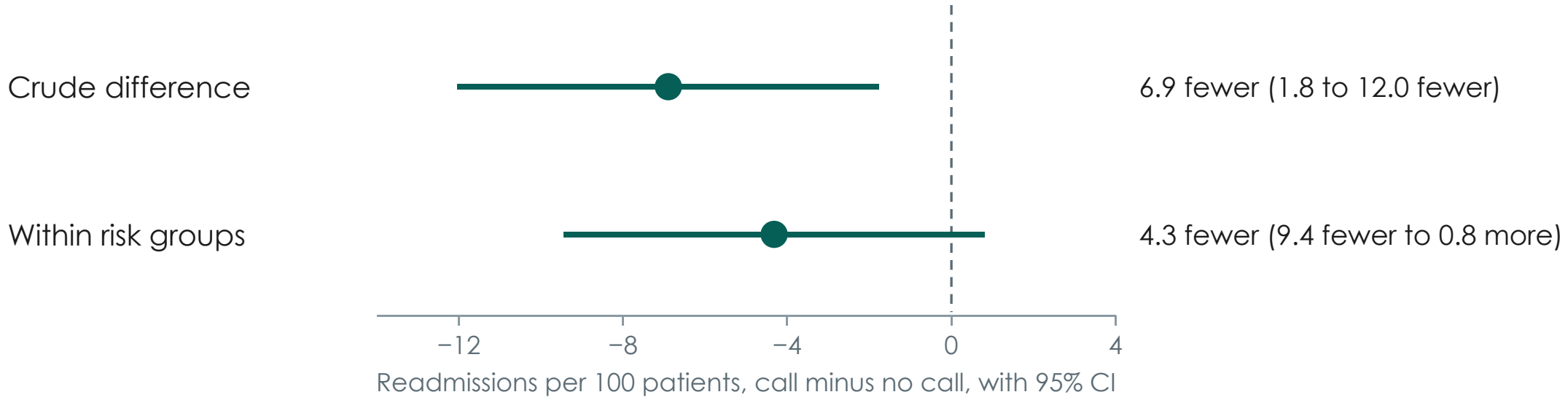
Calls produce a massive drop in readmissions!



Fictional teaching data, not a real study.

AFTER

Readmissions were 6.9 per 100 lower with a call, but the data allow a wide range



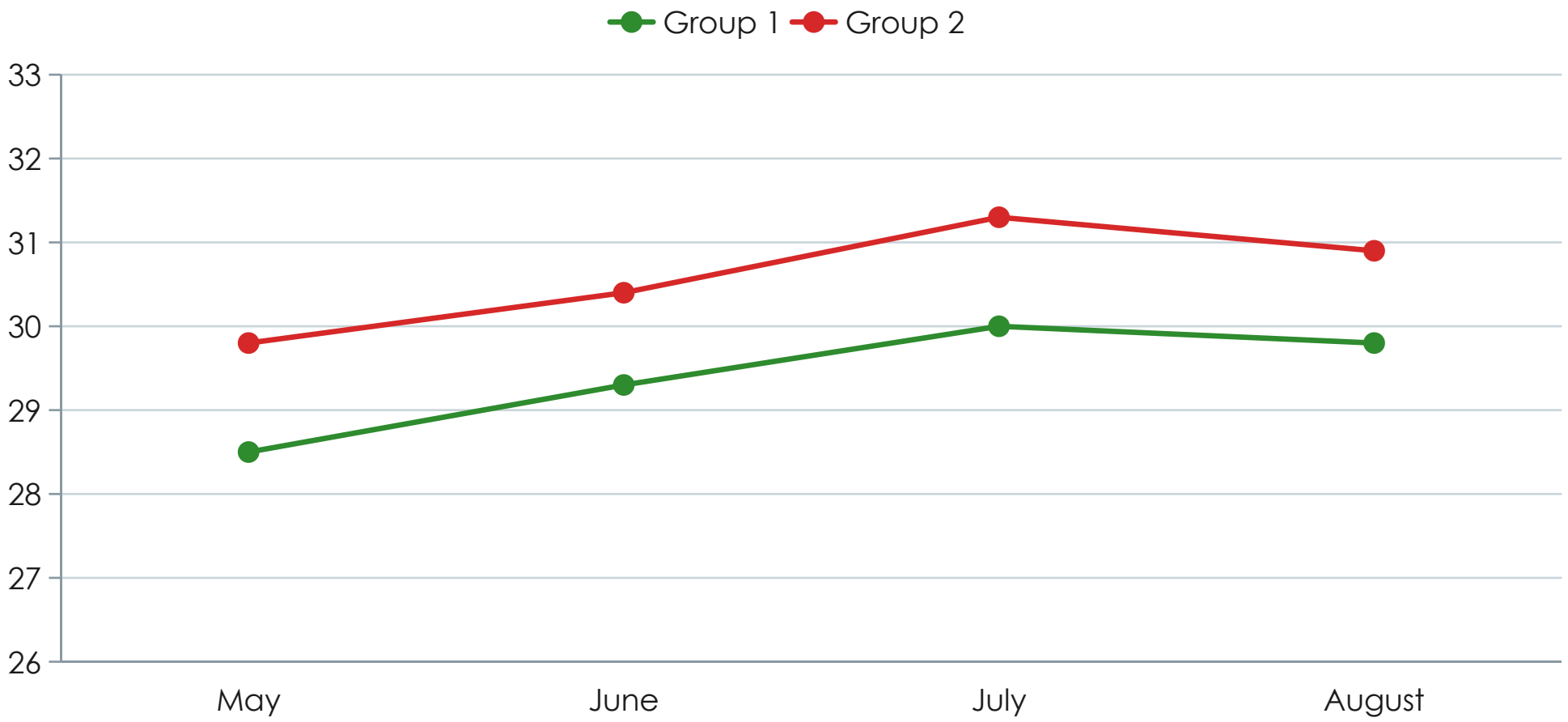
Risk of readmission: 60 of 412 (14.6%) with a call, 94 of 438 (21.5%) without.

The dot uses position, so the axis can focus on the range of the differences.

Fictional teaching data, not a real study.

BEFORE

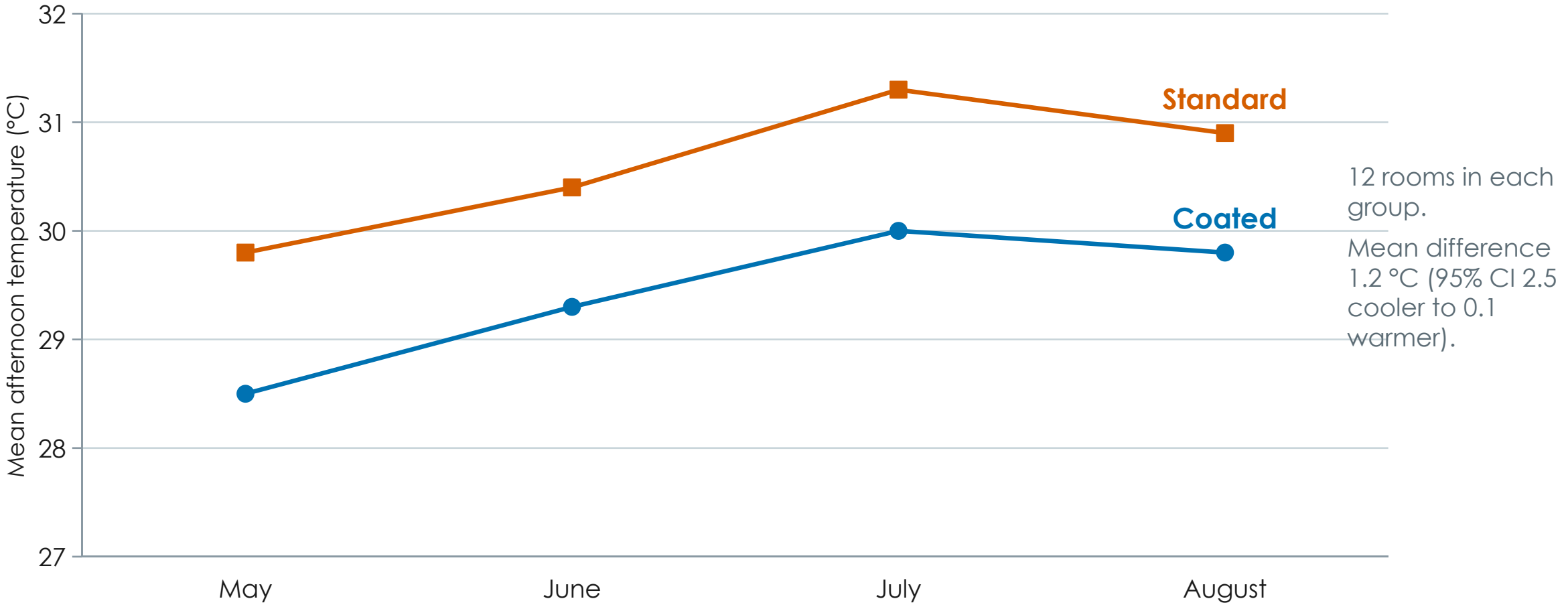
Temperature comparison by month



Fictional teaching data, not a real study.

AFTER

Coated classrooms averaged 1.2 °C cooler, but the interval includes no difference



12 rooms in each group.

Mean difference 1.2 °C (95% CI 2.5 cooler to 0.1 warmer).

Fictional teaching data, not a real study.

BEFORE

Conclusions

- Nurse telephone follow-up is a promising intervention
- Further research is needed
- Limitations of this study include its observational design
- The team would like to acknowledge everybody who helped

Thank you! Any questions?

Fictional teaching data, not a real study.

AFTER

Recommendation: test calls on two wards for six months before funding full roll-out

What we know

4.3 fewer readmissions per 100 within risk groups.

The data fit 9.4 fewer to 0.8 more.

What it costs

About 137 nurse hours for 412 calls at 20 minutes each.

What we do not know

Whether calls caused the difference, because reachable patients were healthier.

Decision needed today

Approve a staggered six-month trial on two wards, so that allocation no longer depends on who can be reached. Decide on full roll-out from its results.

Fictional teaching data, not a real study.

Critique the next slide, then redesign it

Activity, 13 minutes. Worksheet 4. Critique in pairs (3), redesign alone (9).

Critique, 3 minutes

What is the claim?

What is the evidence?

What would a colour-blind viewer lose?

What does 'NS' hide?

Redesign, 9 minutes

Write the headline as a sentence

Choose one visual for the evidence

Keep the denominators

Keep the uncertainty

Paper is fine. A clear sketch beats a polished slide with no claim.

BEFORE

Subgroup analysis

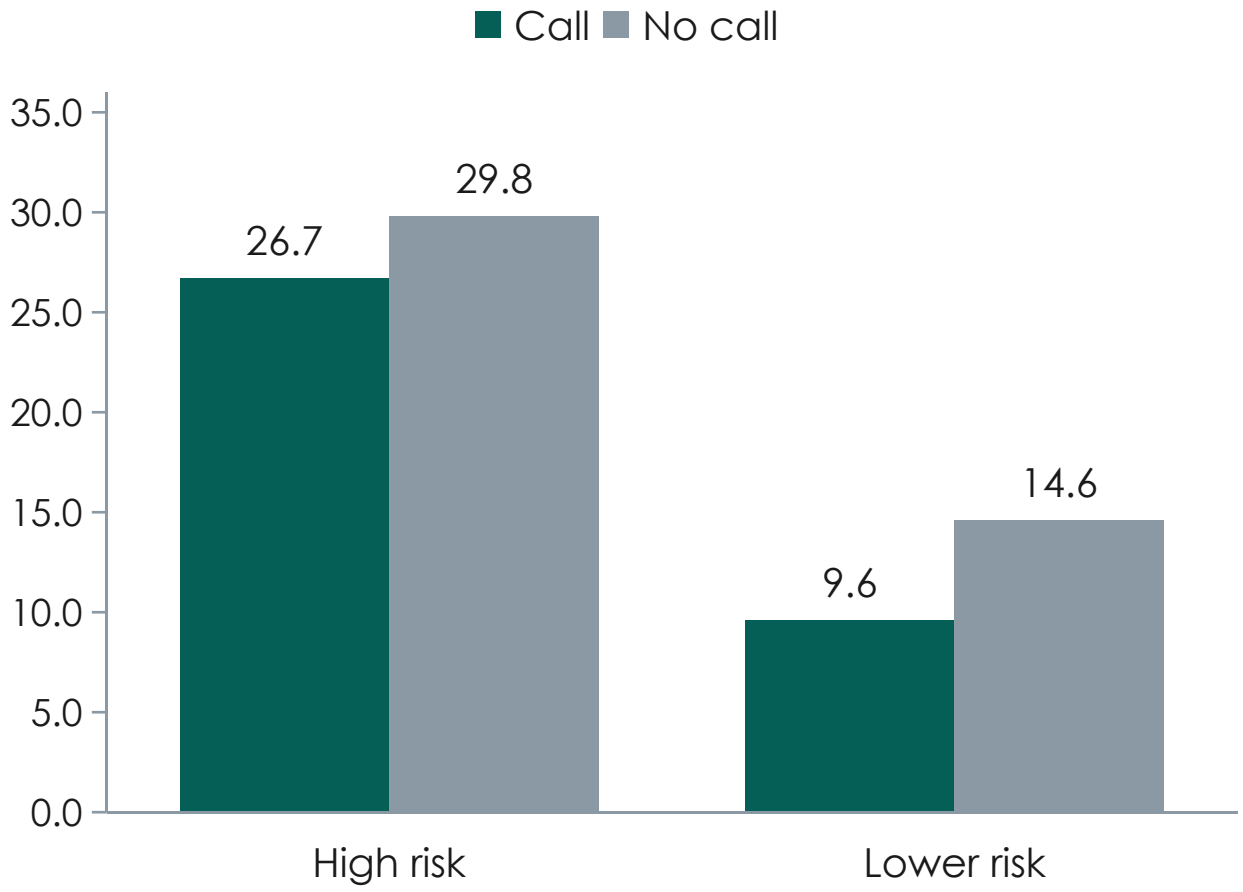
Stratum	Call events	Call n	Call %	No-call events	No-call n	No-call %	p
High	32	120	26.7	59	198	29.8	NS
Lower	28	292	9.6	35	240	14.6	NS

- Green = good, red = bad
- No significant difference in either stratum, so calls had no effect in subgroups
- However the overall analysis was significant
- See table for further details

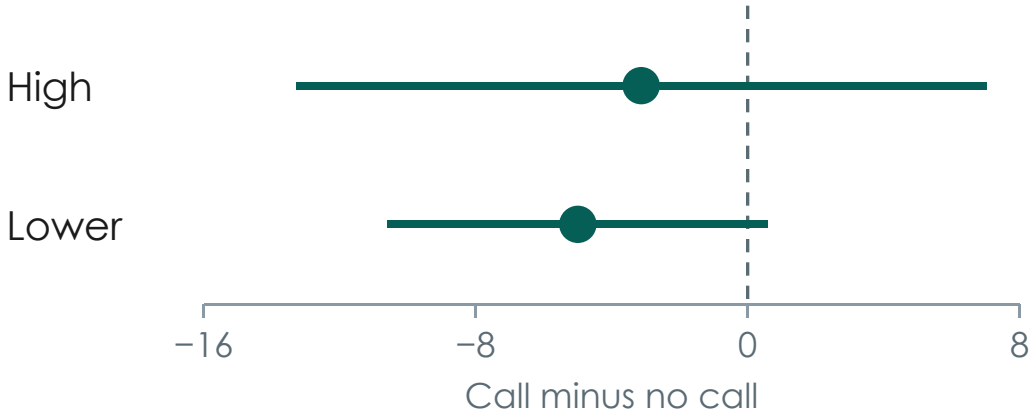
Fictional teaching data, not a real study.

AFTER

In both risk groups fewer patients who got a call were readmitted, and the intervals are wide



Difference per 100 (95% CI)



High risk: 32 of 120 v 59 of 198.

Lower risk: 28 of 292 v 35 of 240.

Too few patients to separate these groups.

Fictional teaching data, not a real study.

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Delivery skills

Module 5: 10 minutes

Explain every figure in four moves

1 What it shows	"Each dot is the difference in readmissions, per hundred patients."
2 Where to look	"Look at the top line, and the dashed line at zero."
3 What you see	"With a call there were about 7 fewer. The interval stays below zero."
4 Why it matters	"So there is a signal. The next line shows how much survives comparing like with like."

Fictional teaching data, not a real study. Teaching recommendation, drawing on W3C Making Events Accessible and the assertion-evidence tutorial.

Clear speech is about being understood, not sounding like someone else

Accent

How your speech differs from what a listener expects. Not a fault, and never scored here.

Intelligibility

How much of your message the listener actually recovers.

Comprehensibility

How easy your speech is to follow. Pace, pauses and structure all help.

Habits that help every listener

Pause after each key number.

Say long numbers slowly and show them.

Signpost: "Two results. First..."

Stress the word that carries the point.

Rehearse your technical terms aloud.

Research finding: Munro & Derwing 1995 (abstract, small study). Review: Derwing & Munro 2009. Guidance: W3C Making Events Accessible.

Rehearse against the real clock and decide in advance what to cut

Say it aloud →	Time it →	Cut →	Run for a listener
Full run, standing, with the clicker. Silent reading does not count.	Use the real limit. Note the time at each section.	Remove before you speed up.	Use the feedback prompts. Then revise.

Mark your slides before you rehearse

Must keep	Should keep	Can cut
Question, main result, limit, conclusion	Second result, method detail	Extra examples, background already known

If you run over on the day, jump to your conclusion. Never skip it.

Professional advice: Bourne 2007, Naegle 2021, Penn State. We found no study of rehearsal timing itself.

Say your opening and one transition aloud

Activity, 5 minutes. Pairs. Each person has 2 minutes, plus 30 seconds to swap.

Opening

State the problem and your question, in under 30 seconds.

Transition

Join two storyboard cards with one sentence.

Partner

Say one thing you remember.

"That is the overall result. But were these patients really alike?"

Useful feedback says what happened to you as a listener

What was the main message? Say it back in your own words.

Where did you stop following? Name the moment, the term or the number.

What one change would help most? Choose one. Make it specific.

Off limits: accent, looks, personality, or "you should be more confident".

Review: Hattie & Timperley 2007, Double et al. 2020. Both extrapolated to adult talks.

Each trio has three rounds of exactly six minutes

Activity, 18 minutes. Trios. Three rounds. Use the rehearsal checklist and Worksheet 5.

Speak	Write	Discuss	Redo
2 min	1 min	2 min	1 min

Speak: explain your redesigned slide with the four moves. The timekeeper stops you at 2:00.

Write: both listeners complete the feedback prompts in silence.

Discuss: message, where they stopped following, one change.

Redo: repeat your weakest part once.

Then rotate speaker, timekeeper and listener.

What changed between your first attempt and your redo?

Where did your two minutes go?

What did a listener hear differently from what you meant?

What will you cut first?

Listen, repeat, answer briefly, then check

Listen to the end → **Repeat or rephrase** → **Answer briefly** → **Check it landed**

Question type	What to do	You could say
Critical	Find the valid part and answer that	"That is a fair concern. Reachability may explain part of this."
Unclear	Check what they mean	"Do you mean the crude result or the stratified one?"
Out of scope	Say so and offer a route	"We did not study cost per patient. I can follow up afterwards."
Unknown	Say so plainly	"I do not know. I will check and reply by email."

Professional and teaching guidance: Streeter & Miller 2011, Dunphy 2007, NIU, W3C Making Events Accessible.

Practise three kinds of question in three-minute rounds

Activity, 9 minutes. Trios. Three rounds of 3 minutes. Question cards on Worksheet 7.

Question

30 seconds: a partner reads a card.

Answer

60 seconds: listen, repeat, answer, check.

Feedback

90 seconds: was it answered? Was the uncertainty honest?

Useful phrases

"Let me check I understood: you are asking about..."

"I do not know. The data cannot answer that."

"That is outside this study. I can follow up afterwards."

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Accuracy and responsible use

Module 6: 7 minutes

Simplify the words, and keep the claim the same size

Check	Original	Over-simplified	Honest and simple
Size	6.9 points lower	Far fewer	About 7 fewer in every 100
Population	Adults, medical wards	All patients	Adults on medical wards
Direction of claim	Associated with	Calls cut readmissions	Patients with a call came back less often
Certainty	9.4 fewer to 0.8 more	Proven	We cannot yet rule out no benefit

Fictional teaching data, not a real study. Research finding: Sumner et al. 2014 (observational, UK, 2011).

Credit the people, and protect the people in your data

Credit

Cite figures and data you did not make

Name collaborators and their roles

Ask before you acknowledge someone

Use your field's authorship criteria

Confidentiality

No names, initials or record numbers

Rare case + ward + date can identify

Small cells can expose a patient or clinician

Coded data are usually not anonymous

Find your organisation's rule. Thresholds differ, and legal rules differ by country.

Policy: ICMJE; HHS de-identification guidance (US); GDPR Recital 26 (EU). Guidance: Public Health Scotland protocol.

AI can help you edit and rehearse; you verify every fact, reference and claim

Useful for

Tightening sentences you wrote

Finding hard-to-say phrases

Suggesting likely questions

Timing a script

Check by hand

Every number against your table

Every reference: open it

Every verb: did the claim grow?

Every summary against the source

Do not

Paste patient or confidential data into tools your organisation has not approved

List AI as an author

Skip disclosure when required

In one 2025 study, most language models tested widened scientific claims when summarising, and asking for accuracy did not help. A polished sentence can still be an overstated one.

Research findings: Walters & Wilder 2023, Peters & Chin-Yee 2025. Policy: ICMJE 2026, COPE. Check your journal and institution.

Check your own slide against your source

Activity, 3 minutes. Alone. Tick each item on your redesigned slide.

- [ ] Every number matches the source table
- [ ] Denominators and units are shown
- [ ] Uncertainty is shown
- [ ] The verb fits the design
- [ ] Headline is no stronger than the data
- [ ] No identifiable person
- [ ] Sources and collaborators credited
- [ ] AI use disclosed, if required

Choose one revision, and book one rehearsal

One revision

What will you change first?

One rehearsal

With whom, and on what date?

One check

Which claim will you verify?

Write them on your personal plan in the workbook, then complete the course feedback form.

Keep the rubric, rehearsal checklist and references for your next talk.



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Backup: the feedback rubric at a glance

Criterion	1 Not yet	2 Developing	3 Secure
Message	No clear claim	Claim present, drifts	One claim, listener repeats it
Structure	List of topics	One gap in the chain	Question to implication connects
Writing	Vague or overstated	Clear in most places	Clear, no stronger than evidence
Evidence	Numbers without context	Most context given	Size, denominator, uncertainty
Visual clarity	Hard to read	Readable, cluttered	Readable, claim visible
Timing	Over by more than a quarter	Rushed ending	Within the limit, even pace
Questions	Bluffs or argues	Partly answers	Answers, states limits

Backup: how the CareCall numbers were calculated

Quantity	Calculation	Result
Risk with a call	60 / 412	14.6%
Risk with no call	94 / 438	21.5%
Crude difference	Wald 95% CI	6.9 lower (1.8 to 12.0)
Within risk groups	Mantel-Haenszel, Greenland-Robins variance	4.3 lower (9.4 lower to 0.8 higher)
Emergency visits	40/412 v 47/438	1.0 lower (5.1 lower to 3.0 higher)
Nurse hours	412 calls x 20 min / 60	About 137 hours

Fictional teaching data, not a real study.

Backup: where the guidance in this workshop comes from

Type	Examples	How to read it
Research finding	Garner & Alley 2013; Akl 2011; Pandey 2015; Sumner 2014	One study or review. Check its setting and limits.
Review or consensus	Altman & Bland 1995; ASA 2016; Greenland 2016	Expert summary, usually reliable on principle.
Standard or policy	WCAG 2.2; ICMJE 2026; COPE; institutional defence rules	Binding only where it applies.
Professional guidance	Bourne 2007; Naegle 2021; SAMPL; CDC Index	Experience-based. Useful, not proven.
Our recommendation	Chain, cut list, four moves, rubric	Teaching devices, labelled as ours.